

CO1-AT2- Agile Sprint Planning & User Story Workshop

Problem Statement: Smart Museum Artifact Preservation Platform

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1. Project Vision & Stakeholders

Vision: Develop a smart platform that continuously monitors museum artifacts, predicts preservation risks, manages digital archives, and notifies museum authorities to ensure long-term conservation.

Objectives:

- Monitor artifact conditions.
- Detect environmental changes.
- Predict preservation risks.
- Manage digital archives.
- Generate conservation reports.
- Notify museum authorities.

Stakeholders:

- Museum Administrator
- Curator
- Conservation Officer
- Security Team
- Visitors (indirect)
- System Administrator

2. Epics & User Stories

Epic	User Story
Artifact Monitoring	As a curator, I want to monitor artifact conditions so that I can preserve valuable objects.
Environment Monitoring	As a conservation officer, I want to detect environmental changes so that damage can be prevented.
Risk Prediction	As an administrator, I want AI-based preservation risk prediction so that I can take preventive actions.
Digital Archive	As a curator, I want to manage digital archives so that artifact records are easily accessible.
Reporting & Alerts	As an administrator, I want conservation reports and alerts so that I receive timely notifications.

3. Acceptance Criteria

- Given an artifact, when sensor data is updated, then the latest condition is displayed.
- Given abnormal temperature/humidity, when thresholds are exceeded, then an alert is generated.
- Given historical data, when prediction runs, then a preservation risk level is shown.
- Given a new artifact, when details are submitted, then they are stored in the archive.
- Given monthly data, when report generation is requested, then a downloadable report is created.

4. Product Backlog

Priority	Feature	Reason
High	Artifact Monitoring	Core functionality
High	Environmental Detection	Prevents damage
High	Risk Prediction	Early warning
Medium	Digital Archive	Documentation
Medium	Reports & Notifications	Operational support

5. Sprint Backlog (Sprint 1)

Tasks:

- Design database for artifacts.
- Develop artifact monitoring module.
- Implement environmental sensor data processing.
- Create alert notification module.
- Test monitoring dashboard.

6. Story Point Estimation (Fibonacci)

User Story	Points	Justification
Monitor artifacts	5	Moderate complexity
Detect environment	8	Sensor integration
Predict risks	13	AI model development
Digital archive	5	CRUD operations
Reports & alerts	3	Simple reporting

7. Sprint Goal & Deliverables

Sprint Goal: Deliver the first working version of the Smart Museum Artifact Preservation Platform with artifact monitoring, environmental monitoring, and alert features.

Expected Deliverables:

- Artifact monitoring dashboard.
- Environmental monitoring module.
- Basic risk prediction interface.
- Digital archive prototype.
- Alert notification system.

Code of Conduct:

I V.Rakshana(192425137) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.Rakshana